

# AE-382: Synthesis of [Ru(CO)(OH)<sub>2</sub>(PNP*t*Bu)] using Ag<sub>2</sub>O, 5 h

Date: 2024-11-13

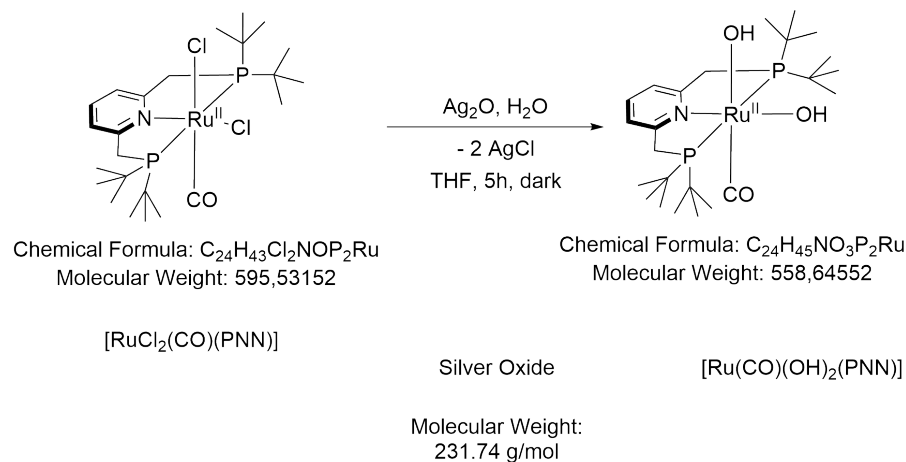
Tags: Dihydroxo Ag<sub>2</sub>O PNP  
P(*t*Bu)N(*py*)P(*t*Bu) [Ru(CO)(OH)<sub>2</sub>(PNP)]  
NMR ESI-MS AE MS reference analytics  
1H 31P Elemental Analysis

Category: Two-photon

Status: Done

Created by: Alexander Eith

## Reaction scheme/sample structure



## Literature/reference experiments

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Literature          | <a href="https://doi.org/10.1039/D1EE01053K">https://doi.org/10.1039/D1EE01053K</a>                                                                                                                                                                                                                                                                                                                                                         |
| Reproduction        | /                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Related experiments | Experiment - AE-KRA-204: Synthesis of [Ru(CO)(OH) <sub>2</sub> (PNP <i>t</i> Bu)] using Ag <sub>2</sub> O, 5 h<br>Experiment - AE-238: Synthesis of [Ru(CO)(OH) <sub>2</sub> (PNP <i>t</i> Bu)] using Ag <sub>2</sub> O, 5 h<br>Experiment - AE-252: Synthesis of [Ru(CO)(OH) <sub>2</sub> (PNP <i>t</i> Bu)] using Ag <sub>2</sub> O, 5 h<br>AE-310: Synthesis of [Ru(CO)(OH) <sub>2</sub> (PNP <i>t</i> Bu)] using Ag <sub>2</sub> O, 5 h |

## Reagents

| Name                                      | CAS-Number/Experiment-Number                                                                                                                                      | Amount [mmol] | Equivalents | Mass <sub>theo</sub> [mg] | Mass <sub>exp</sub> [mg] | Molar mass [g/mol] | Volume <sub>theo</sub> [ml] | Volume <sub>exp</sub> [ml] | Density [g/mL] |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------|---------------------------|--------------------------|--------------------|-----------------------------|----------------------------|----------------|
| [RuCl <sub>2</sub> (CO)(PNP <i>t</i> Bu)] | Experiment - AE-261: Synthesis of [RuCl <sub>2</sub> (CO)(PNP)] with 2,6-Bis(tert-butyl)phosphinomethylpyridine via [RuCl <sub>2</sub> (CO)(p-cymene)] (AE-261-1) | 0.827         | 1.00        | -                         | 492.57                   | 595.53             | /                           | /                          | /              |
| Ag <sub>2</sub> O                         | 20667-12-3                                                                                                                                                        | 5.078         | 6.14        | 1176.88                   | 1176.70                  | 231.74             | /                           | /                          | /              |
| H <sub>2</sub> O                          | Experiment - KRA-104: Degassing of Milli-Q water                                                                                                                  | 193.54        | 234         | 3487                      | /                        | 18.02              | 3.49                        | 3.6                        | 0.998          |
| THF                                       | Experiment - KRA-080: Degassing of THF                                                                                                                            | /             | /           | /                         | /                        | /                  | 25                          | 25                         | /              |
| n-Heptane                                 | Experiment - AE-121: Degassing of n-Heptane                                                                                                                       | /             | /           | /                         | /                        | /                  | /                           | 20                         | /              |

## Work-up Reagents

| Name      | CAS Number /                                | Inventory number | Mass <sub>exp</sub> [g] | Volume [ml] | Concentration [M] |
|-----------|---------------------------------------------|------------------|-------------------------|-------------|-------------------|
| THF       | Experiment - KRA-080: Degassing of THF      | /                | /                       | 25 +20      | /                 |
| n-heptane | Experiment - AE-121: Degassing of n-Heptane | /                | /                       | 59 + 55     | /                 |
| MeCN      | AE-152: Degassing of MeCN                   | /                | /                       | 6           | /                 |

## Procedure/observations

All steps, unless mentioned otherwise, were carried out under argon using standard Schlenk technique or in an argon filled glovebox. (see Protocol [Protocol - Schlenk Technique](#))

Due to the instability of Ag<sub>2</sub>O and Dihydroxo to the light, the procedure was done mostly in the dark, by either wrapping the flasks containing either or both in Aluminium foil or by tuning off the light in the lab and fumehood

| Date  | Time          | Step                                                                                                                                       | Observations                                                                |
|-------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 14.11 | 9:10          | A 50mL- Schlenk flask was prepared, by heating under vacuum                                                                                |                                                                             |
|       | 9:30          | [RuCl <sub>2</sub> (CO)(PNP)] and Ag <sub>2</sub> O was weighted in air into the reaction schlenk flask and the flask was set under argon. |                                                                             |
|       |               | The light in the fumehood was turned out                                                                                                   |                                                                             |
|       | 10:10         | THF (25 mL) was added to the reaction schlenk flask.                                                                                       |                                                                             |
|       | 10:15         | Water (3.6 mL) was added to the reaction schlenk flask                                                                                     |                                                                             |
|       | 10:15 - 15:15 | The reaction mixture was stirred for 5 h at 700 rpm. During this the flask was wrapped with aluminium foil                                 | <a href="#">Start of reaction.jpg</a><br><a href="#">after reaction.jpg</a> |
|       | 15:15 - 15:20 | The reaction mixture was allowed to settle                                                                                                 |                                                                             |

|       |               |                                                                                                                                                                                                                                                                          |                                                                               |
|-------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
|       | 15:20 - 15:50 | The mixture was filtered through a pore 4 frit according to <a href="#">Protocol - Filtration with frit technique</a> into a 100 mL-Schlenk                                                                                                                              |                                                                               |
|       | 15:50 - 16:05 | The reaction flask was washed with MeCN (2 * 3 mL)                                                                                                                                                                                                                       | 1st washing: yellowish liquid phase<br>2nd washing: nearly clear liquid phase |
|       | 16:05 - 17:05 | The solvents of the filtrate were removed under reduced pressure.<br>• When approx. 5 mL were left the water bath was removed and the mixture was allowed to freeze (at approx. 16:20)<br>• At 16:45 the water bath was added again to slowly melt the remaining solvent | This procedure avoids bubbling up of the mixture                              |
|       |               | The solid was stored in the dark at - 20 °C                                                                                                                                                                                                                              | Dark brownish solid<br><a href="#">Solid after night.jpg</a>                  |
| 15.11 | 10:20 - 10:30 | To the solid THF (25 mL) was added and the obtained mixture was stirred and shaken strongly for approx. 7 min                                                                                                                                                            | <a href="#">after adding THF.jpg</a>                                          |
|       | 10:30 - 10:37 | n-heptane (50 mL) was added to the mixture                                                                                                                                                                                                                               | <a href="#">after adding n-hept.jpg</a>                                       |
|       | 11:17 - 12:00 | The mixture was filtered through a pore 3 frit according to <a href="#">Protocol - Filtration with frit technique</a> into a 100 mL-Schlenk                                                                                                                              |                                                                               |
|       | 12:03 - 12:10 | The solid was washed with n-heptane (3 * 3 mL)                                                                                                                                                                                                                           |                                                                               |
|       | 12:13 - 13:17 | The solid was dried under reduced pressure                                                                                                                                                                                                                               |                                                                               |
|       | 13:30 - 13:40 | The solid was transferred into a 2 mL vial in a 10 mL Schlenk flask                                                                                                                                                                                                      | Some material was lost, since it stuck strongly to any surface                |
|       |               | During this an aliquote (approx. 5 mL) was transferred into another 10 mL Schlenk flask                                                                                                                                                                                  |                                                                               |
|       | 13:45 - 14:35 | Both flasks were dried under reduced pressure                                                                                                                                                                                                                            |                                                                               |
|       |               | Both flask were set under argon                                                                                                                                                                                                                                          |                                                                               |
|       | 14:50         | <a href="#">From the aliquote n NMR sample was prepared in DMSO-d6 according to Protocol - Preparation of NMR Sample</a>                                                                                                                                                 | <b>AE-382-1</b> <a href="#">AE-382-1.jpg</a>                                  |
|       |               | The remaining solid was stored in the dark under argon at -20 °C                                                                                                                                                                                                         | <b>AE-382-1</b><br><a href="#">Product.jpg</a>                                |

|       |               |                                                                                                                                                                                                                                                                                                                                                             |                                                                             |
|-------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 04.12 |               | Further workup, since in <sup>1</sup> H NMR some more sideproducts are observed (with all remaining <b>AE-382-1</b> )                                                                                                                                                                                                                                       |                                                                             |
|       |               | A 100 mL flask was prepared                                                                                                                                                                                                                                                                                                                                 |                                                                             |
|       |               | The vial containing <b>AE-382-1</b> was added to the flask is Ar counterflow                                                                                                                                                                                                                                                                                |                                                                             |
|       | 16:05         | To the 100 mL flask THF (20 mL) was added                                                                                                                                                                                                                                                                                                                   | <a href="#">in THF.jpg</a> dark suspension                                  |
|       |               | It was tried to remove th vial without sucess                                                                                                                                                                                                                                                                                                               |                                                                             |
|       | 16:07 - 16:12 | The flask was shaken to get all solids out of the vial                                                                                                                                                                                                                                                                                                      |                                                                             |
|       | 16:12 - 16:23 | The flask was stirred                                                                                                                                                                                                                                                                                                                                       | brownish suspension                                                         |
|       | 16:23 - 16:30 | To the flask n-heptane (52 mL) was added                                                                                                                                                                                                                                                                                                                    | <a href="#">after n-hept.jpg</a> orange solution, brownish.gray solid       |
|       | 16:30 - 16:32 | The suspension was stirred                                                                                                                                                                                                                                                                                                                                  |                                                                             |
|       |               | The flask was stored in the dark in <a href="#">Equipment - Freezer Lab E004 CEEC II</a> under Ar                                                                                                                                                                                                                                                           |                                                                             |
| 05.12 |               | The obtained suspension was warmed to rt                                                                                                                                                                                                                                                                                                                    | <a href="#">before filt.jpg</a> orange solution, brownish.gray solid        |
|       | 10:40 - 14:30 | The suspension was filtered through a pore 4 filter frit                                                                                                                                                                                                                                                                                                    | Very slow filtration, filtration was stopped after approx. 40 mL of solvent |
|       | 14:30 - 14:50 | The obtained filter residue was washed with n-Heptane (1 * 1.5 mL)                                                                                                                                                                                                                                                                                          | <a href="#">washing of filterate on 1st frit.jpg</a>                        |
|       | 14:50 - 14:58 | The obtained solid ( <b>AE-382-2</b> ) was dried under reduced pressure                                                                                                                                                                                                                                                                                     |                                                                             |
|       | 14:53         | The filtration setup was changed to a pore 3 frit. Therefore the new frit was connected with a "Vakuumvorstoß", closed and set under inert conditons. Afterwards the cannula, still conected to the flask containing the suspesion, was attached in Ar counterflow. Then the reciving flask was added under Ar counterflow and the filtration was continued |                                                                             |
|       | 14:56 - 15:13 | The remaining suspension was filtered through a pore 3 filter frit                                                                                                                                                                                                                                                                                          |                                                                             |
|       | 15:15 - 15:20 | The obtained residue ( <b>AE-382-3</b> ) was washed with n-heptane (3 * 1 mL)                                                                                                                                                                                                                                                                               |                                                                             |

|       |               |                                                                                                                                                                                                |                                                          |
|-------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
|       | 15:30 - 17:05 | AE-382-2 and -3 were dried under reduced pressure                                                                                                                                              |                                                          |
|       |               | Both were stored on the frit in the dark under Ar in <a href="#">Equipment - Freezer Lab E004 CEEC II</a>                                                                                      |                                                          |
| 09.12 | 10:40         | <b>AE-382-3:</b> The solid was transferred into a 1.5 mL screw cap vial in a 10 mL Schlenk flask. during this approx. 5 mg were added to another 10 mL Schlenk flask to prepare an NMR sample. |                                                          |
|       |               | Both flasks were set under argon and stored in <a href="#">Equipment - Freezer Lab E004 CEEC II</a> in the dark under Ar                                                                       |                                                          |
| 12.12 | 11:05         | To the flask containing the NMR sample DMSO-d6 (0.75 mL) was added                                                                                                                             | <a href="#">-3 in DMSO-d6.jpg</a> brownish solution      |
|       | 11:05 - 11:10 | The obtained solution was stirred                                                                                                                                                              |                                                          |
|       | 11:10         | <a href="#">An NMR sample was prepared according to Protocol - Preparation of NMR Sample</a>                                                                                                   | <a href="#">AE-382-3.jpg</a> brownish solution           |
| 15.01 | 10:05         | <b>AE-382-2:</b> The solid was transferred into a 1.5 mL screw cap vial in a 10 mL Schlenk flask. during this approx. 4 mg were added to another 10 mL Schlenk flask to prepare an NMR sample. |                                                          |
|       |               | Both flasks were set under argon and <b>AE-382-2</b> was stored in <a href="#">Equipment - Freezer Lab E004 CEEC II</a> in the dark under Ar                                                   |                                                          |
|       | 10:25         | To the flask containing the NMR sample MeCN-d3 (0.65 mL) was added                                                                                                                             |                                                          |
|       |               | <a href="#">An NMR sample was prepared according to Protocol - Preparation of NMR Sample</a>                                                                                                   | <a href="#">AE-382-2.jpg</a>                             |
| 13.03 |               | A 10 mL and a 5 mL Schlenk flask were prepared, each containing a 1.5 mL screw cap vial without lid                                                                                            |                                                          |
|       | 13:45         | An sample for an elemental analysis was prepared by weighing in 3.75 mg of <b>AE-382-2</b> into the vial of the 10 mL Schlenk flask and setting it under argon afterwards.                     | <b>AE-382-EA</b> (AE382a on sheet, since max. 6 letters) |
|       | 13:50         | An sample for MS was prepared. Therefore 1.62 mg of <b>AE-382-2</b> was transferred into the vial of the 5 mL Schlenk flask in air. Afterwards the flask was set under argon.                  | <b>AE-382-MS</b>                                         |

|       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |
|-------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|       | 14:00 | <p>The sample was submitted to Nicole Fritz. Further work done by Nicole:</p> <p>Rough description of sample handling: The flask was opened to air and a small amount (approx. 0.1 mg) was taken out and diluted in acetonitril (ACN) 3 times. After taking out the small amount the flask was closed again. Afterwards the sample was injected into the MS device. Before and after the measurment a calibration measurement was performed. (<b>AE-382-MS ACN_1</b>)</p> <p>After approx. 10 min a sample of the same solution was measured (<b>AE-382-MS ACN_1_10min</b>)</p> <p>A new sample was prepared the same way as the first sample (<b>AE-382-MS ACN_2</b>)</p> <p>A new sample was prepared analog to the first sample, but the small amount taken out was first dissolved in water (approx. 0.5 mL) and than diluted with ACN (<b>AE-382-MS H2O_ACN</b>)</p> |  |
| 31.03 | /     | The elemental analysis of <b>AE-382-EA</b> was measured by Sandra Köhn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |

## Analysis

| Date  | Time  | Sample name           | Analysis method      | Used device    | Solvent    | Raw Data                 | Processed Data                                           | Used for plotting | Comparative Data                                                                                                                     | Interpretation                                                                                                                                                                                                      |
|-------|-------|-----------------------|----------------------|----------------|------------|--------------------------|----------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18.11 | 13:21 | AE-382-1              | NMR 1H, 31P{1H}quant | IAAC 400 MHz I | DMSO-d6    | AE-382-1.zip             | AE-382-1_10.nmrium                                       | /                 | AE-310: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h                                                                           | As expected. Only minor impurities                                                                                                                                                                                  |
| 12.12 | /     | AE-382-3              | NMR 1H, 31P{1H}quant | IAAC 400 MHz I | DMSO-d6    | AE-382-3.zip             | 2024-12-13_12-58-51.zip_AE-382-3_10.nmrium               | /                 | /                                                                                                                                    | As expected. Only minor impurities                                                                                                                                                                                  |
| 15.01 | 17:31 | AE-382-2              | NMR 1H, 31P{1H}quant | IAAC 400 MHz I | MeCN-d3    | AE-382-2.zip             | AE-382-2_10.nmrium                                       | /                 | /                                                                                                                                    | As expected. Only minor impurities                                                                                                                                                                                  |
| 13.03 | 14:10 | AE-382-MS ACN_1       | ESI-MS               | ESI-TOF-Bruker | MeCN       | AE-382-MS-ACN_1.xy       | AE-382-2-MS-ACN_1.txt<br>AE-382-MS-ACN_1.png             | ESI-MS.py         | C24H43NO3P2RuCO.csv<br>calculated via<br><a href="https://www.envipat.eawag.ch/index.php">https://www.envipat.eawag.ch/index.php</a> | Very weak singal which could be M+H, but significant signals at higher mass, full interpretation pending<br>Main signal could be from carbonate forming and removing both OH groups through addition of CO2         |
| 13.03 | 14:20 | AE-382-MS ACN_1_10min | ESI-MS               | ESI-TOF-Bruker | MeCN       | AE-382-MS-ACN_1_10min.xy | AE-382-2-MS-ACN_1_10min.txt<br>AE-382-MS-ACN_1_10min.png | ESI-MS.py         | C24H43NO3P2RuCO.csv<br>calculated via<br><a href="https://www.envipat.eawag.ch/index.php">https://www.envipat.eawag.ch/index.php</a> | Significantly different pattern -> most likely decomposition                                                                                                                                                        |
| 13.03 | 14:30 | AE-382-MS ACN_2       | ESI-MS               | ESI-TOF-Bruker | MeCN       | AE-382-MS-ACN_2.xy       | AE-382-2-MS-H2O_ACN.txt<br>AE-382-MS-ACN_2.png           | ESI-MS.py         | C24H43NO3P2RuCO.csv<br>calculated via<br><a href="https://www.envipat.eawag.ch/index.php">https://www.envipat.eawag.ch/index.php</a> | Very weak singal which could be M+H, but significant signals at higher mass, full interpretation pending<br>Weaker signal than ACN_1. signal at 627 less pronounced. Main signal at 586 u (delta to expected: 28 u) |
| 13.03 | 14:35 | AE-382-MS H2O_ACN     | ESI-MS               | ESI-TOF-Bruker | MeCN + H2O | AE-382-MS-H2O_ACN.xy     | AE-382-2-MS-ACN_2.txt<br>AE-382-MS-H2O_ACN.png           | ESI-MS.py         | C24H43NO3P2RuCO.csv<br>calculated via<br><a href="https://www.envipat.eawag.ch/index.php">https://www.envipat.eawag.ch/index.php</a> | Very weak singal which could be M+H, but significant signals at higher mass, full interpretation pending<br>Some new, high mass peaks                                                                               |
| 31.03 | /     | AE-382-EA             | elemental analysis   | ?              | /          | AE-382-EA.jpg            | C: 49.78 %<br>H: 6.94 %<br>N: 2.45 %                     | /                 | C: 51.6 %<br>H: 8.12 %<br>N: 2.51 %                                                                                                  | C and H contant to low by 1.8 pp for C and 1.2 pp for H. N content fits roughly                                                                                                                                     |

## Product characterization and results

| Sample   | Mass [mg] | Purity | Mass <sub>pure</sub> [mg] | Amount [μmol] | Yield [%] | Description                                                              | Storage location |
|----------|-----------|--------|---------------------------|---------------|-----------|--------------------------------------------------------------------------|------------------|
| AE-382-1 | 149.79    | > 96 % | /                         | 0.268         | 32        | grayish olive-ochre very fine, sticky solid, <a href="#">Product.jpg</a> | used             |
| AE-382-3 | 40.87     | > 96 % | /                         | /             | /         | grayish olive-ochre very fine, sticky solid                              | freezer drawer A |
| AE-382-2 | 23.87     | > 96 % | /                         | /             | /         | grayish olive-ochre very fine, sticky solid                              | freezer drawer A |

MS fits to +CO -HO.

## Linked experiments

- [AEI-010: Synthesis of \[Ru\(CO\)\(OH\)2\(PNN\)\] using Ag2O](#)

- [JSC-KS-04: Degassing of n-Heptane](#)

- [AEI-024: Synthesis of \[Ru\(CO\)\(OH\)2\(NNN\)\] with 2,6-bis\(N,N-diethylaminomethyl\) using Ag2O](#)

- [AEI-026: Capillary production](#)

- [AEI-048: Synthesis of \[Ru\(CO\)\(OH\)2\(NNN\)\] with 2,6-bis\(N,N-diisopropylaminomethyl\) using Ag2O](#)

- [AEI-064: Synthesis of \[Ru\(CO\)\(OH\)2\(PNP\)\] with 2,6-bis\(P,P-di\(tertbutyl\)phosphinomethyl\) using Ag2O](#)

- [AEI-073: Synthesis of \[Ru\(CO\)\(OH\)2\(PNP\)\] with 2,6-bis\(P,P-di\(tertbutyl\)phosphinomethyl\) using Ag2O](#)

- [AE-101: Synthesis of \[RuCl2\(CO\)\(PNP\)\] with 2,6-Bis\(\(tert-butyl\)phosphinomethyl\)pyridine via \[RuCl2\(CO\)\(p-cymene\)\]](#)

- [AE-102: Degassing of THF](#)

- [AE-105: Synthesis of \[Ru\(CO\)\(OH\)2\(PNP\)\] with 2,6-bis\(P,P-di\(tertbutyl\)phosphinomethyl\) using Ag2O](#)

- [AE-108: Synthesis of \[RuCl2\(CO\)\(PNP\)\] with 2,6-Bis\(\(tert-butyl\)phosphinomethyl\)pyridine via \[RuCl2\(CO\)\(p-cymene\)\]](#)

- AE-114: Synthesis of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$  with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using  $\text{Ag}_2\text{O}$
- AE-121: Degassing of n-Heptane
- AE-145: Synthesis of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$  with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using  $\text{Ag}_2\text{O}$
- AE-152: Degassing of MeCN
- AE-157: Synthesis of  $[\text{RuCl}_2(\text{CO})(\text{PNP})]$  with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via  $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$
- AE-167: Irradiation of PhPDA (365 nm + white LED, gas phase measurement), AE-151, 1.5 mg/mg SDS, 2 mg/mL PhPDA
- AE-198: Synthesis of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$  with  $\text{Ag}_2\text{O}$  with different reaction times (1, 2, 5, 24 h)
- AE-KRA-204: Synthesis of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$  using  $\text{Ag}_2\text{O}$ , 5 h
- AE-208: Degassing of Milli-Q water
- KRA-080: Degassing of THF
- KRA-104: Degassing of Milli-Q water
- AE-230: Synthesis of  $[\text{RuCl}_2(\text{CO})(\text{PNP})]$  with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via  $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$
- AE-232: Synthesis of  $[\text{RuCl}_2(\text{CO})(\text{PNP})]$  with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via  $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$
- AE-238: Synthesis of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$  using  $\text{Ag}_2\text{O}$ , 5 h
- AE-246: Radiation of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$  (AE-238, 97 % pure) 365 nm and white
- AE-252: Synthesis of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$  using  $\text{Ag}_2\text{O}$ , 5 h
- AE-261: Synthesis of  $[\text{RuCl}_2(\text{CO})(\text{PNP})]$  with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via  $[\text{RuCl}_2(\text{CO})(\text{p-cymene})]$
- AE-281: Stability test of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$  (AE-252) under air and afterwards irradiation with 365 nm and white
- AE-310: Synthesis of  $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$  using  $\text{Ag}_2\text{O}$ , 5 h



## Linked resources

Equipment - [Freezer Lab E004 CEEC II](#)

Protocol - [Preparation of NMR Sample](#)

Protocol - [Filtration with frit technique](#)

Protocol - [Schlenk Technique](#)

Protocol - [Preparation of capillary for H<sub>2</sub>O-NMR](#)

## Attached files

AE-382-EA.jpg

sha256: 4f383ed83f0808ff6e6830a4ea048a0522ef4c7b5a5ec1265b0a43b53c06d46f



C24H43NO3P2RuCO.csv

sha256: e324012daf18017a2ef1d1d4b926e79458494cebb8e25ae6d29361227fe719fd

ESI-MS.py

sha256: c5a21c33581840e3dd1f8fc8697c254996c7dc42ae6121e877fb052a499b9e58

AE-382-2-MS-H<sub>2</sub>O\_ACN.txt

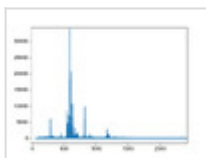
sha256: abf859520850a5b424900e4f39f7c0f01901a7becbe0999eb2ca0f9091f33221

AE-382-2-MS-ACN\_2.txt

sha256: ab0cfb93750befe6ae85bff2c668ed879a7a4319ff120611cdbe4db5275721bb

AE-382-MS-H<sub>2</sub>O\_ACN.png

sha256: f40efc75601edc7db4941c312b70af01d2188cad98a60f521b4b24477eb6243



AE-382-2-MS-ACN\_1.txt

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AE-382-MS-ACN\_1\_10min.xy

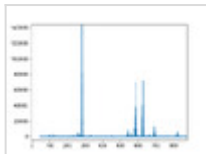
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AE-382-MS-ACN\_1.xy

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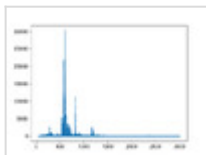
AE-382-MS-ACN\_1.png

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AE-382-MS-ACN\_1\_10min.png

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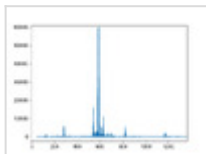


AE-382-2-MS-ACN\_1\_10min.txt

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AE-382-MS-ACN\_2.png

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AE-382-MS-ACN\_2.xy

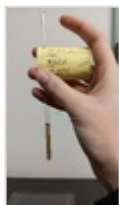
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AE-382-MS-H2O\_ACN.xy

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AE-382-2.jpg

sha256: 38f8235762d0898057456031830bf15ea796c9815115412088ace58e177c60e8



AE-382-2\_10.nmrium

sha256: dd67522ebeda93741474bc5723fed107f200e7270475dce713868e9864864ca6

AE-382-2.zip

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AE-382-3.zip

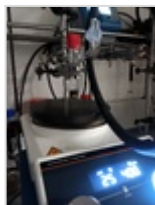
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2024-12-13\_12-58-51.zip\_AE-382-3\_10.nmrium

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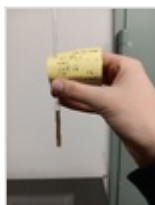
3-in-DMSO-d6.jpg

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AE-382-3.jpg

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before-filt.jpg

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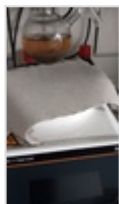
in-THF.jpg

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after-n-hept.jpg

sha256: c206c51b6df0041594e2dc1483adc2ef125658cb21b9e4714e1f87e7e125d400



washing-of-filtrate-on-1st-frit.jpg

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AE-353.png

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AE-382-1\_10.nmrium

sha256: 105e1c8e653ddd70ce6e280e15fc0654b8480fc2a1e0c5d596bb9c24f0865d41

AE-382-1.zip

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after-reaction.jpg

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Product.jpg

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after-adding-n-hept.jpg

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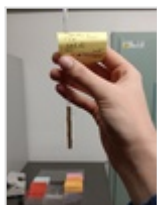
Start-of-reaction.jpg

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AE-382-1.jpg

sha256: 555cb7dce37d606c6ff3860d0c15a178bf5a4872e5d66b505f55f82813c9046e



after-adding-THF.jpg

sha256: 2b3b591b6e78597aac125ed4fb6cb1ca996867bd973e1e4fbb7831c338bea000



Solid-after-night.jpg

sha256: 80072f5c2fb15e55a786a9b31a9883328501a0e964a7fc1639d3ae5a6beb43e4



AE-20421.cdxml

sha256: 70d2a67f6ae7477a3c31ec8e776e001181c617b18e0d276cd2995bdbdfdb2ca2

## Comment

On 2024-12-13 13:41:11 Jacob Schneidewind wrote:

Notes:

\* adding future recommendation to change MeCN purification step (same as AE-302-1)



Unique eLabID: 20241113-5686e080afe9a3b141d8cfff3b0f1b849e0097d

Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1502>